Fundamentals 18

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1. Consider

$$h(x) = f(x) - \frac{f(b) - f(a)}{g(b) - g(a)}(g(x) - g(a)).$$

We have that $h(a) = f(a)$ and $h(b) = f(b) - (f(b) - f(a)) = f(a)$. Since h is differentiable on (a, b), by Rolle's Theorem, there exists $c \in (a, b)$ where $h'(c) = 0$. This gives us

$$h'(c) = f'(c) - \frac{f(b) - f(a)}{g(b) - g(a)}g'(c) = 0,$$

$$f'(c) = \frac{f(b) - f(a)}{g(b) - g(a)}g'(c), \quad f'(c)(g(b) - g(a)) = g'(c)(f(b) - f(a)),$$

as desired.

2. The proof claims that, for all $k \in [n]$ (where $[n] = \{1, 2, \ldots, n\}$), there is $x_k \in (a, x_{k-1})$ such that $h^{(k)}(x_k) = 0$ because $h^{(k-1)}(a) = h^{(k-1)}(x_{k-1}) = 0$ and $h^{(k-1)}$ is differentiable. But this assumes that $h^{(k-1)}$ is defined at a, which means that $f^{(k-1)}$ is defined at a for all $k \in [n]$. So it is necessary that $f^{(k-1)} : (a, b) \to \mathbb{R}$ should be extendable to continuous functions on at least $[a, b)$, which is implied by $f \in C^{(n-1)}[a, b]$.

3. Let $f(a) = \lim_{x \to a} f(x)$ and $f(b) = \lim_{x \to b} f(x)$. Since limits can have at most one value, $f(a)$ and $f(b)$ can each have at most one value, so f as at most one continuous extension to $[a, b]$.

4. We have that

$$\lim_{x \to c} \frac{f(x) - P_n(x)}{(x - c)^n} = 0, \quad \lim_{x \to c} \frac{f(x) - Q_n(x)}{(x - c)^n} = 0.$$

Subtracting the first limit from the second yields

$$\lim_{x \to c} \frac{(f(x) - Q_n(x)) - (f(x) - P_n(x))}{(x - c)^n} = \lim_{x \to c} \frac{P_n(x) - Q_n(x)}{(x - c)^n} = 0.$$

Now, let $P(x) = P_n(x+c) - Q_n(x+c)$ so that

$$\lim_{x\to c} \frac{P_n(x) - Q_n(x)}{(x-c)^n} = \lim_{x\to 0} \frac{P(x)}{x^n} = 0.$$

Since P_n and Q_n have a degree of n, P has a degree of at most n, so we can write

$$P(x) = \sum_{k=0}^{n} a_k x^k,$$

so

$$\lim_{x\to 0} \frac{P(x)}{x^n} = \lim_{x\to 0} \sum_{k=0}^{n} a_k x^{k-n} = \lim_{x\to 0} \left(\frac{a_0}{x^n} + \cdots + \frac{a_n}{x^0} \right) = 0.$$

If $a_k \neq 0$ for any k from 0 to $n-1$ then the term $\frac{a_k}{x^{n-k}}$ would become unbounded as $x \to 0$, and if $a_n \neq 0$ then the term $\frac{a_n}{x^0}$ would be $\neq 0$, so $\lim_{x\to 0} \frac{P(x)}{x^n} \neq 0$. So we must have that $a_k = 0$ for all k, meaning $P(x) = 0$ for all x. So we have that $P_n(x) - Q_n(x) = P(x-c) = 0$ for all x, so $P_n(x) = Q_n(x)$, so the Taylor polynomial of f is unique.

5. The first four derivatives of sin are

$$\sin^{(0)}(x) = \sin(x), \quad \sin^{(1)}(x) = \cos(x), \quad \sin^{(2)}(x) = -\sin(x), \quad \sin^{(3)}(x) = -\cos(x),$$

before they start looping. So

$$\sum_{n=0}^{5} \frac{f^{(n)}(0)}{n!} x^n = \frac{\sin 0}{0!}x^0 + \frac{\cos 0}{1!}x^1 + \frac{-\sin 0}{2!}x^2 + \frac{-\cos 0}{3!}x^3 + \frac{\sin 0}{4!}x^4 + \frac{\cos 0}{5!}x^5$$

$$= x - \frac{x^3}{6} + \frac{x^5}{120}.$$

6. The first four derivatives of cos are

$$\cos^{(0)}(x) = \cos(x), \quad \cos^{(1)}(x) = -\sin(x), \quad \cos^{(2)}(x) = -\cos(x), \quad \cos^{(3)}(x) = \sin(x),$$

before they start looping. So

$$\sum_{n=0}^{5} \frac{f^{(n)}(0)}{n!} x^n = \frac{\cos 0}{0!}x^0 + \frac{-\sin 0}{1!}x^1 + \frac{-\cos 0}{2!}x^2 + \frac{\sin 0}{3!}x^3 + \frac{\cos 0}{4!}x^4 + \frac{-\sin 0}{5!}x^5$$

$$= 1 - \frac{x^2}{2} + \frac{x^4}{24}.$$

7. We have that $P(x) = \sum_{i=0}^{n} a_i x^i$, and $f^{(i)}(0) = i! \cdot a_i$ for all i, so

$$P_k(x) = \sum_{i=0}^{k} \frac{f^{(i)}(0)}{i!} x^i = \sum_{i=0}^{k} \frac{i! \cdot a_i}{i!} x^i = \sum_{i=0}^{k} a_i x^i.$$

So the remainder is

$$R_k(x) = P(x) - P_k(x) = \sum_{i=k+1}^{n} a_i x^i.$$

This gives us

$$\lim_{x\to 0} \frac{R_k(x)}{x^k} = \lim_{x\to 0} \sum_{i=k+1}^{n} a_i x^{i-k} = \lim_{x\to 0} (a_{k+1}x + \cdots + a_n x^{n-k}) = 0.$$

9a. We claim that $f^{(n)}(x) = P_n\left(\frac{1}{x}\right) e^{-\frac{1}{x^2}}$ for all $n \in \mathbb{N}$ and $x > 0$ where $P_n(x)$ is a polynomial depending on n.

We see that $f^{(0)}(x) = e^{-\frac{1}{x^2}}$ for $x > 0$, so we can let $P_0(x) = 1$. Now, assume that $f^{(n)} = P_n\left(\frac{1}{x}\right) e^{-\frac{1}{x^2}}$ for $x > 0$. We have that

$$\begin{aligned} f^{(n+1)}(x) &= \left(P_n\left(\frac{1}{x}\right) e^{-\frac{1}{x^2}}\right)' = \left(P_n\left(\frac{1}{x}\right)\right)' e^{-\frac{1}{x^2}} + P_n\left(\frac{1}{x}\right)\left(e^{-\frac{1}{x^2}}\right)' \\ &= P_n'\left(\frac{1}{x}\right)\left(\frac{1}{x}\right)' e^{-\frac{1}{x^2}} + P_n\left(\frac{1}{x}\right)\left(e^{-\frac{1}{x^2}}\right)\left(-\frac{1}{x^2}\right)' \\ &\quad \left(-\frac{1}{x^2}\right) P_n'\left(\frac{1}{x}\right) e^{-\frac{1}{x^2}} + \left(\frac{2}{x^3}\right) P_n\left(\frac{1}{x}\right) e^{-\frac{1}{x^2}} \\ &= \left(-\left(\frac{1}{x}\right)^2 P_n'\left(\frac{1}{x}\right) + 2\left(\frac{1}{x}\right)^3 P_n\left(\frac{1}{x}\right)\right) e^{-\frac{1}{x^2}}. \end{aligned}$$

So we can let $P_{n+1}(x) = -x^2 P_n'(x) + 2x^3 P_n(x)$.

Now, we have that

$$\lim_{x\to 0^+} f^{(n)}(x) = \lim_{x\to 0^+} P_n\left(\frac{1}{x}\right) e^{-\frac{1}{x^2}} = \lim_{x\to 0^+} \frac{P_n\left(\frac{1}{x}\right)}{e^{\frac{1}{x^2}}} = \lim_{x\to\infty} \frac{P_n(x)}{e^{x^2}}.$$

Since the exponential term in the denominator grows faster than any polynomial, the limit is equal to 0. Since $f^{(n)}(x) = 0$ for $x \le 0$,

$$\lim_{x\to 0} f^{(n)}(x) = 0 = f^{(n)}(0),$$

so $f^{(n)}(x)$ is continuous at 0 for all n. Also, the function $e^{-\frac{1}{x^2}}$ is C^∞ because it is a composition of functions in C^∞, and 0 is C^∞ as well. This means $f \in C^\infty(\mathbb{R})$.

9b. We have that

$$\sum_{k=0}^{n} \frac{f^{(n)}(0)}{k!} x^k = \sum_{k=0}^{n} \frac{0}{k!} x^k = 0,$$

So $R_n(x) = f(x) - 0 = f(x)$. So the remainder doesn't approach 0 for any $x > 0$.